# 328633 (28)

## BE (6th Semester)
## Examination, April - May, 2016

**(Old Scheme)**

**Medical Electronics**

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*Time Allowed* : 3 hours *Maximum Marks* : 80
*Minimum Pass Marks* : 28

***Note :*** (*i*) Part (*a*) is compulsory and attempt any **two** parts from (*b*), (*c*) and (*d*) of each Unit.

(*ii*) The figures in the right-hand margin indicate marks.

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**Unit - I**

**1.** (*a*) Define the parameters used in medical electronics. 2

(*b*) Mention the components of man-instrument system and also describe their functions. 7

(*c*) Explain physiological system of the body. 7

(*d*) Explain the problems encountered in measuring a living system. 7

*(Turn Over)*

## Unit - II

2. (*a*) Explain the terms bio-electrodes and bio-transducer. 2

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(*b*) How ECG signal is exploited to understand the abnormalities in the function of heart? Describe with examples. 7

(*c*) How biopotentials are generated within the body cells? Explain polarization and depolarization of a cell. 7

(*d*) Explain the heart and cardiovascular system in brief. 7

## Unit - III

3. (*a*) List the various invasive and non-invasive techniques for measurement of blood flow. 2

(*b*) Why Ultrasound is important in medical diagnosis system? Discuss the principles of operation of medical diagnosis using ultrasound. 7

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(*c*) With proper block diagram describe the principle of operation of an electromagnetic blood flow meter. 7

(*d*) How can heart sound be measured electronically? Discuss phonocardiograph with relevant block diagram. 7





## Unit - IV

4. (*a*) What are the elements of intensive care monitoring? 2

(*b*) What is a pacemaker? Classify various types of pacemakers and make a comparison between them. 7

(*c*) How are computers made useful for patient monitoring system? Discuss any one of the cases in which computers are used for patient monitoring? 7

(*d*) Explain the working of a defibrillator. What are its various types? Mention their applications. 7

## Unit - V

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5. (*a*) Define Bio-telemetry. 2

(*b*) Explain the components of Bio-telemetry system and also describe their functions. 7

(*c*) What are the various applications of telemetry for caring a patient? 7

(*d*) Explain the physiological parameters adaptable to bio-telemetry. 7

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